

Detection recall vs. coverage

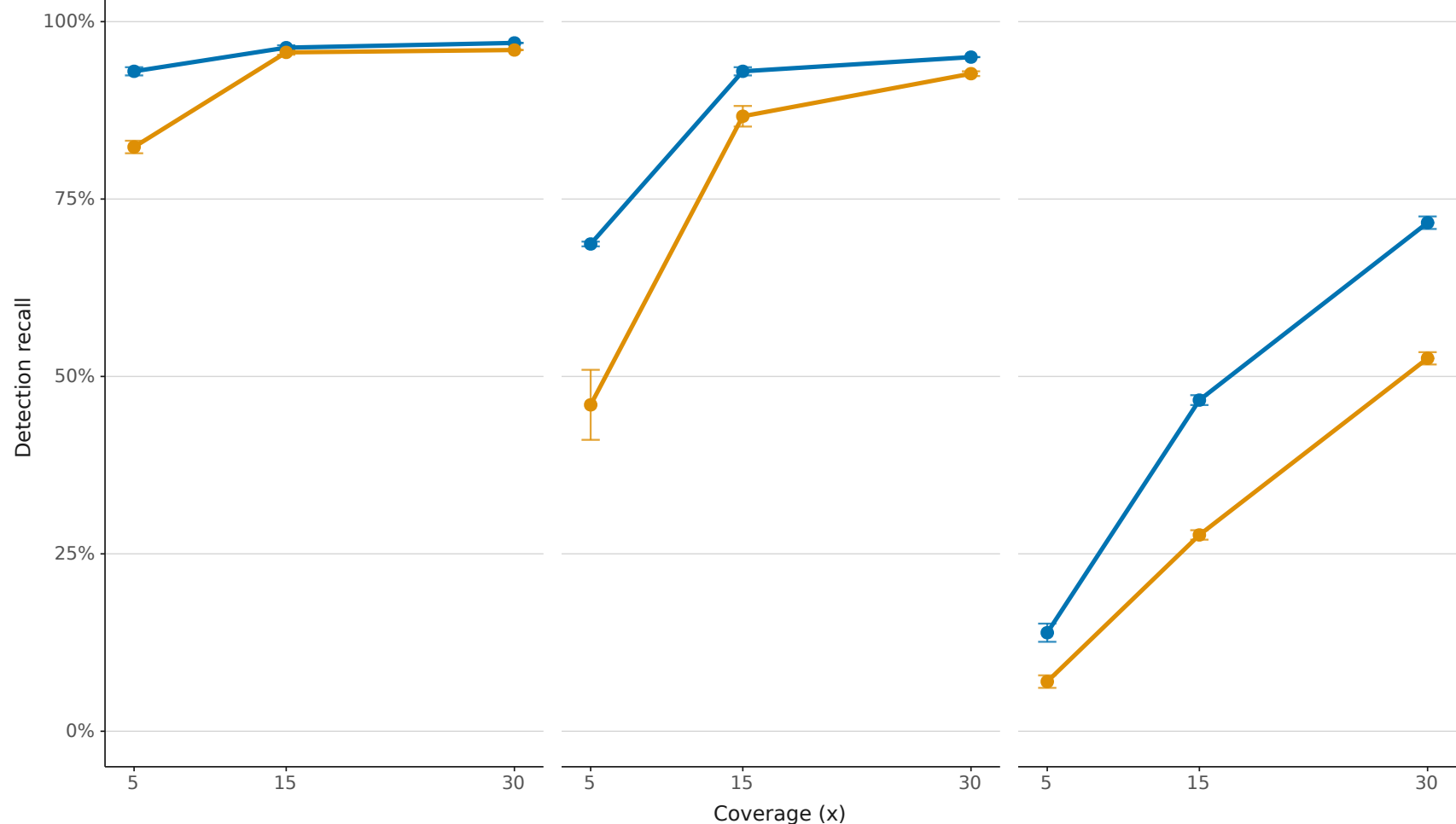
Fraction of simulated insertions recovered (mean $\pm$ SEM, 3 replicates)

RelocaTE2 RelocaTE3

Homozygous

Heterozygous

Somatic



Genotype-status accuracy vs. coverage

Correct hom/het/somatic call among DETECTED insertions (mean $\pm$ SEM)

RelocaTE2 RelocaTE3

Homozygous

Heterozygous

Somatic

Status accuracy | detected

100%
75%
50%
25%
0%

5

15

30

5

15

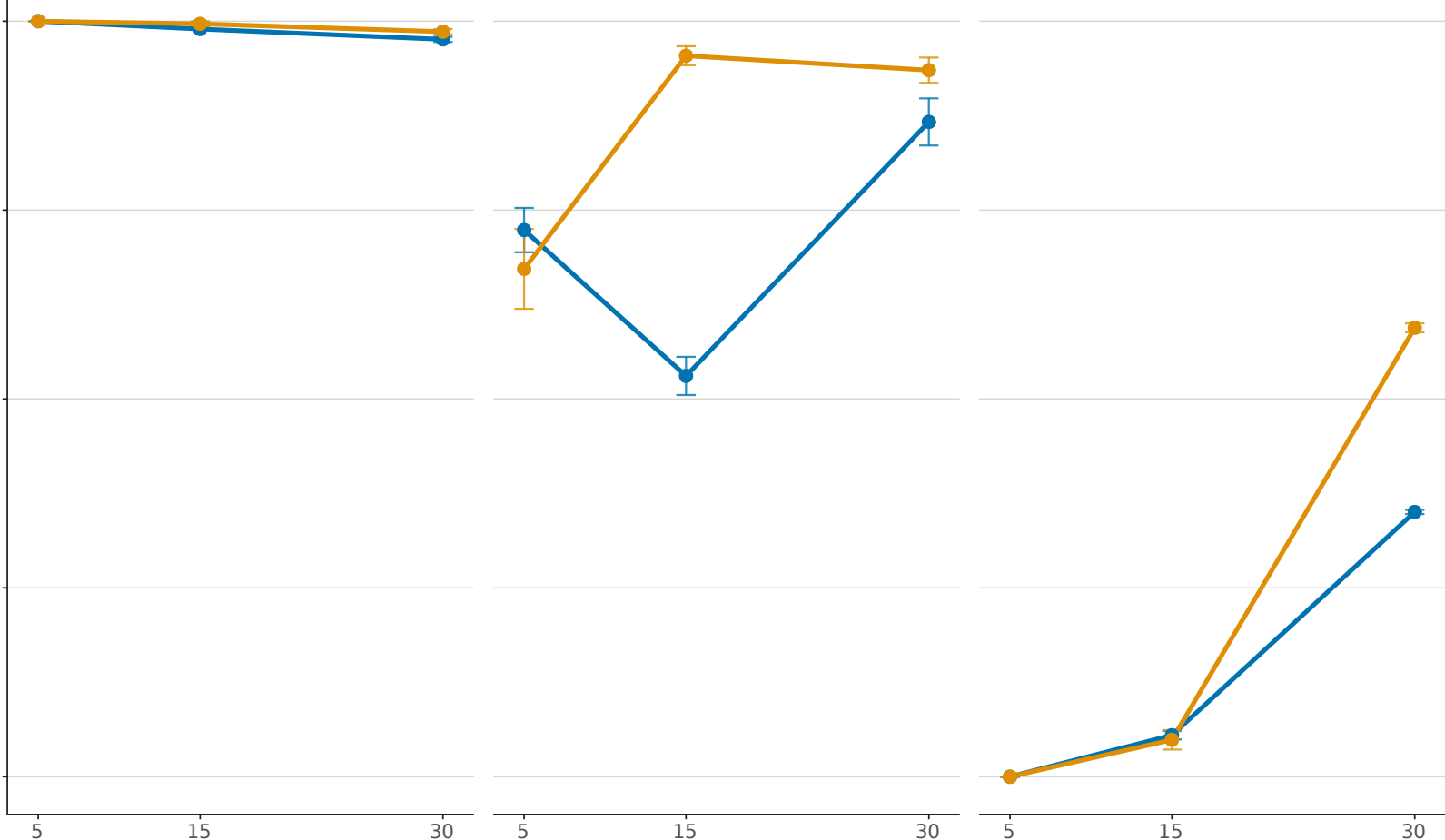
30

5

15

30

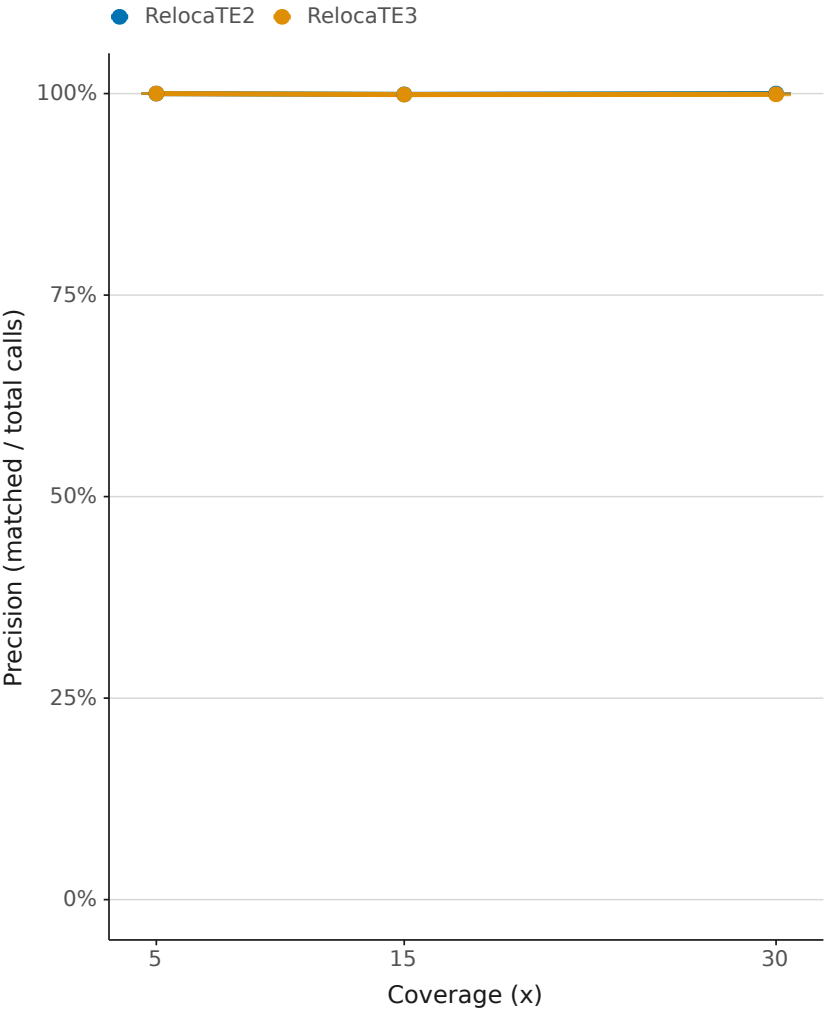
Coverage (x)



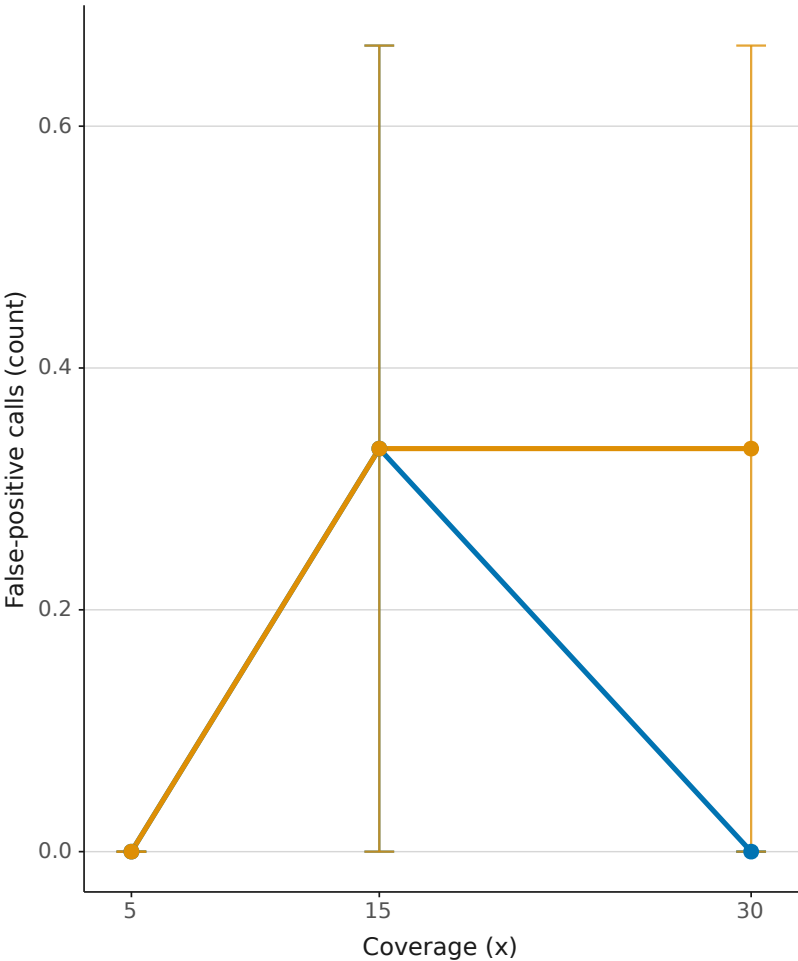
Precision and false positives vs. coverage

RelocaTE precision = fraction of calls matching a true insertion (mean $\pm$ SEM)

Overall precision



False-positive calls



Somatic detection recall by cellular fraction

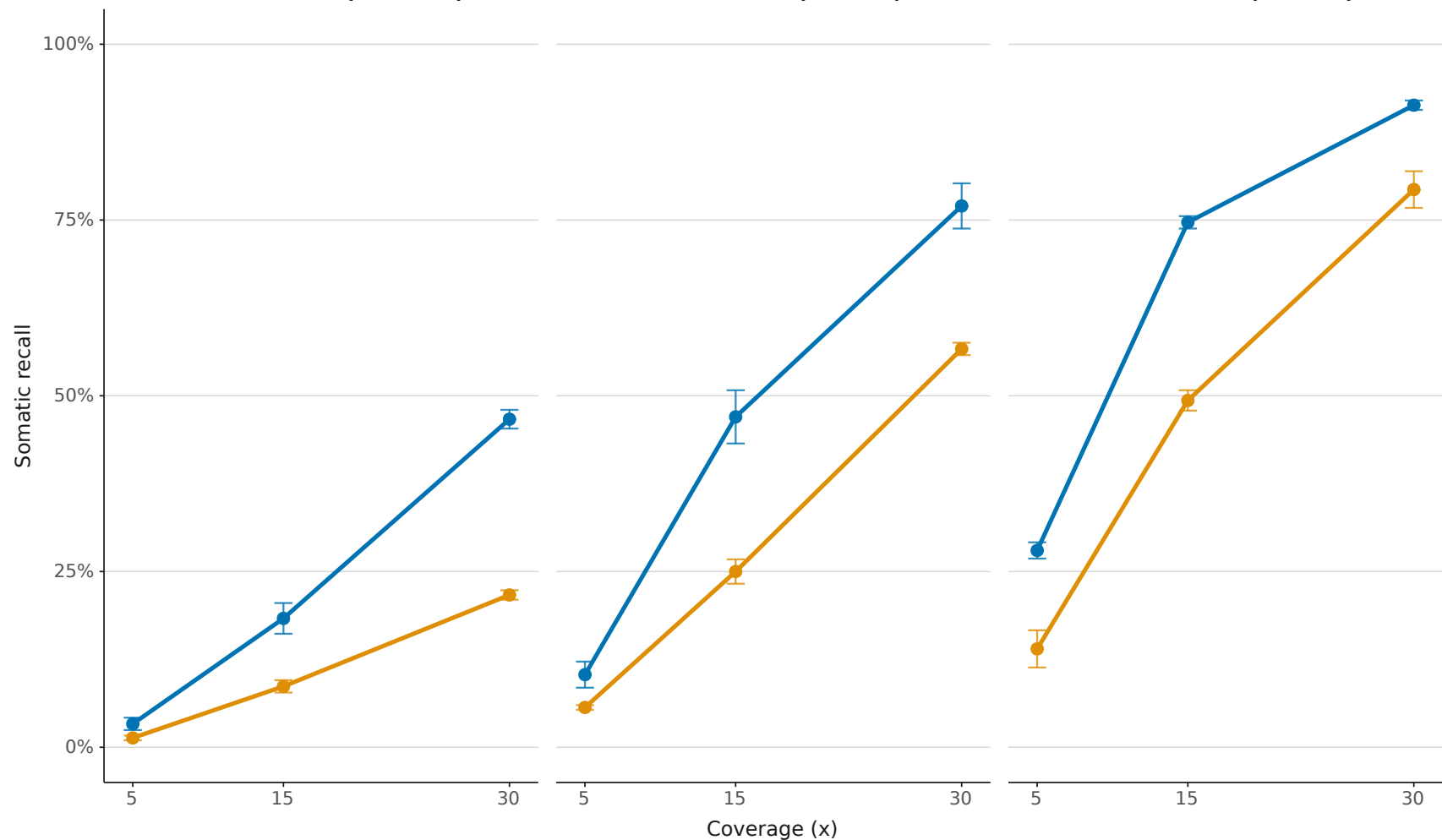
Lower cellular fraction = lower variant allele frequency = harder (mean $\pm$ SEM)

RelocaTE2 RelocaTE3

Cellular fraction 0.1 (VAF 0.05)

Cellular fraction 0.2 (VAF 0.1)

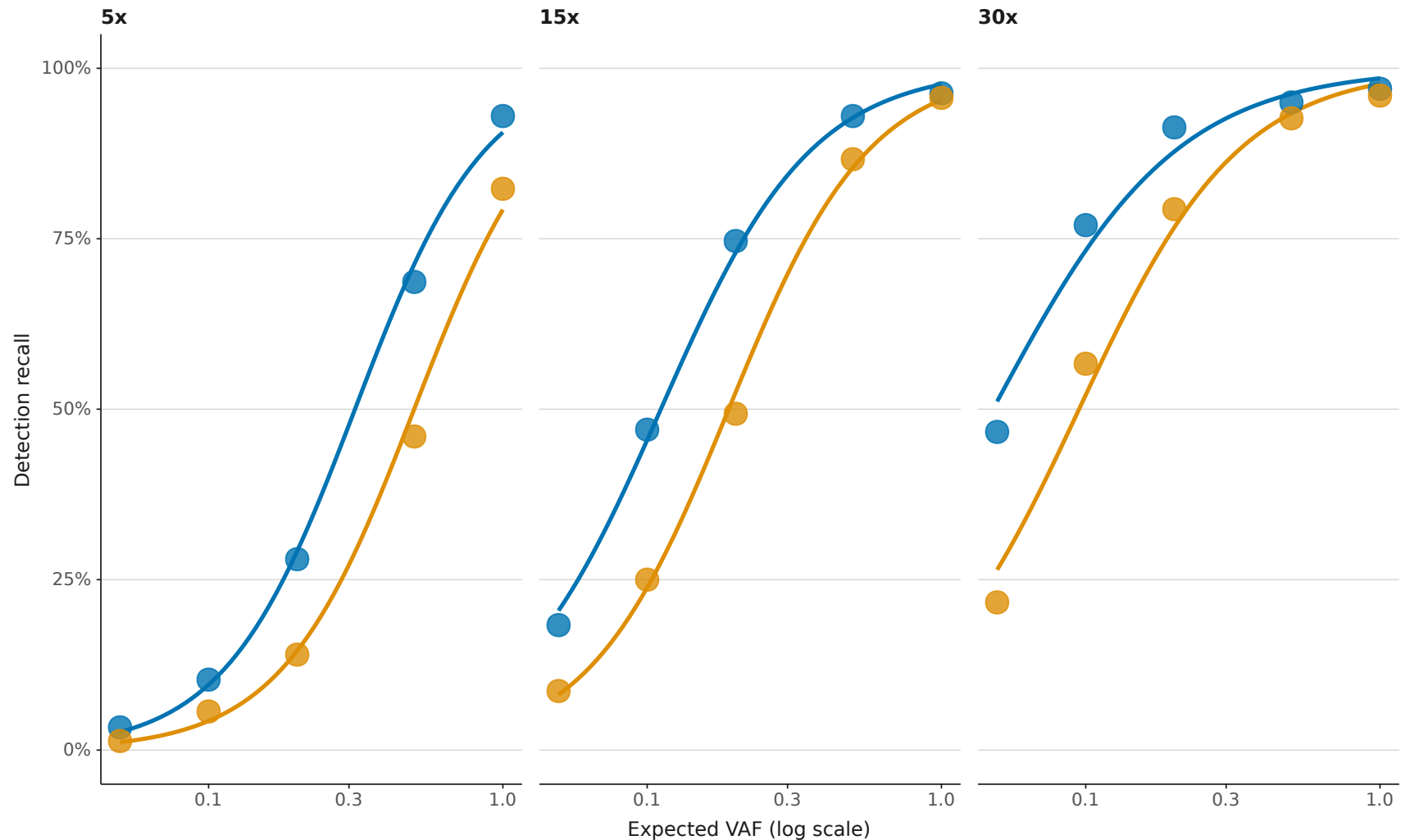
Cellular fraction 0.4 (VAF 0.2)



Detection sensitivity vs variant allele frequency

Logistic fit per caller; VAF at 50% detection = limit of detection (LOD50)

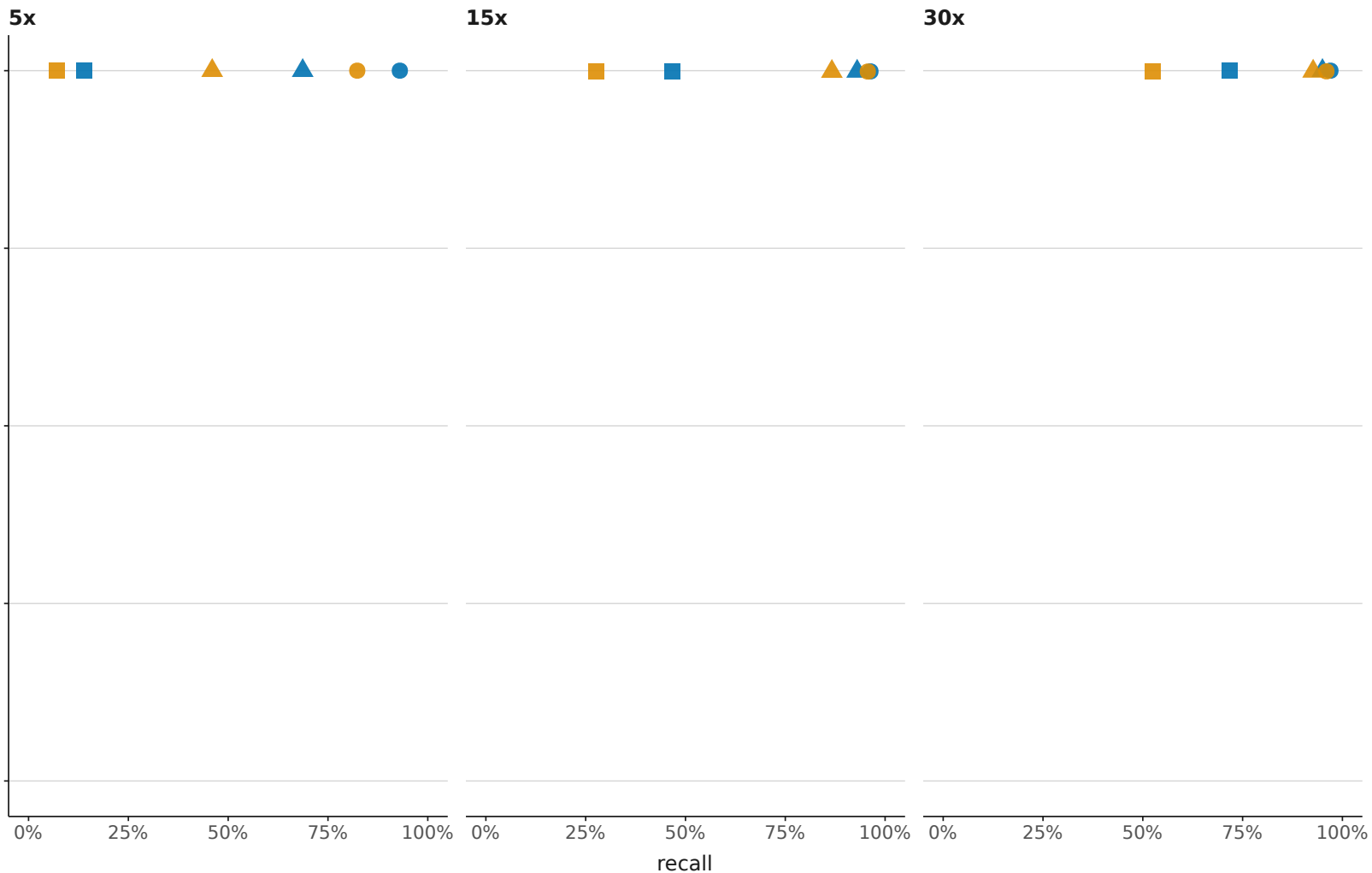
RelocaTE2 RelocaTE3



Precision-recall operating points

Precision = matched / all calls per sample (global denominator); one point per class

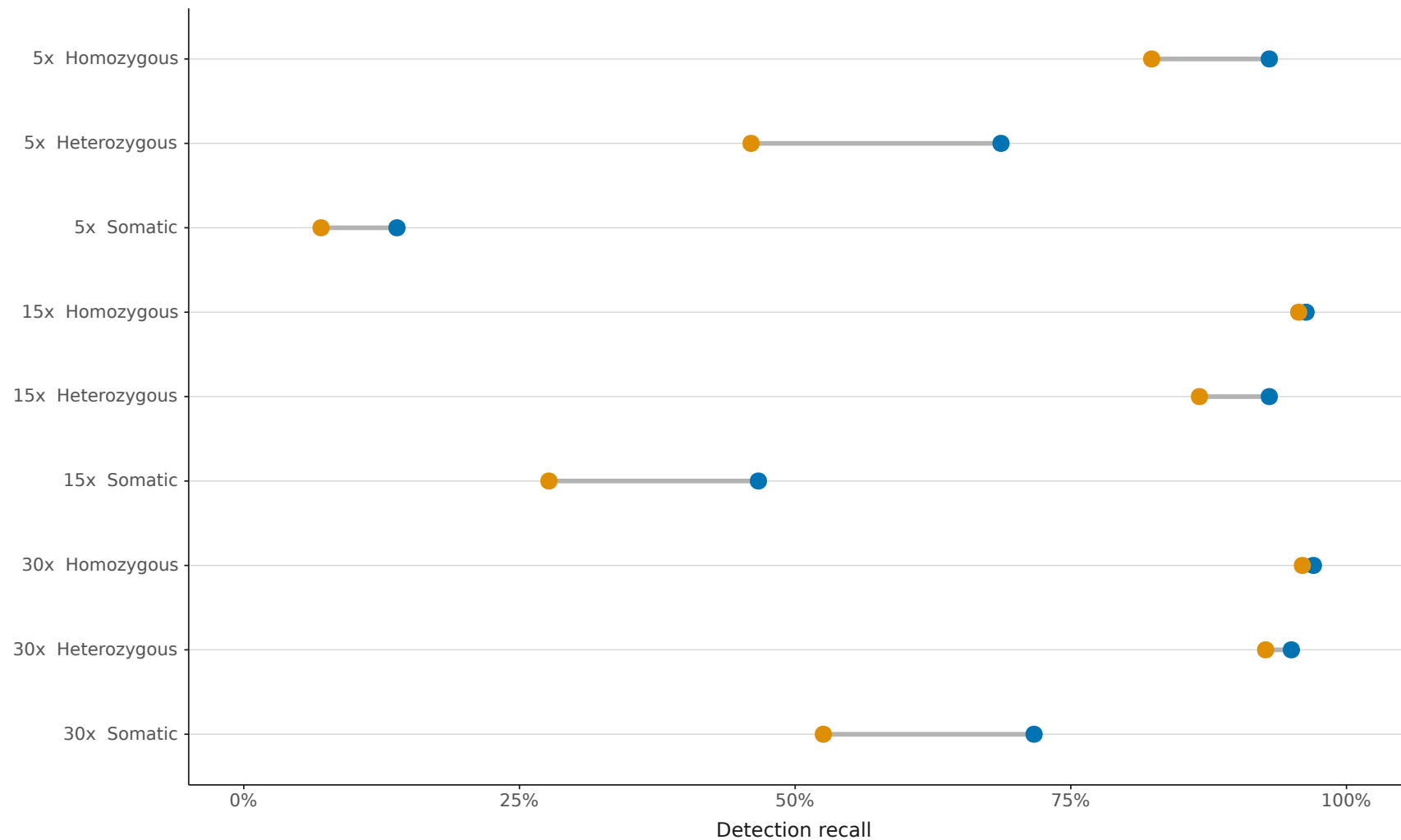
● Homozygous ▲ Heterozygous ■ Somatic ● RelocaTE2 ● RelocaTE3



RelocaTE2 vs RelocaTE3 detection recall

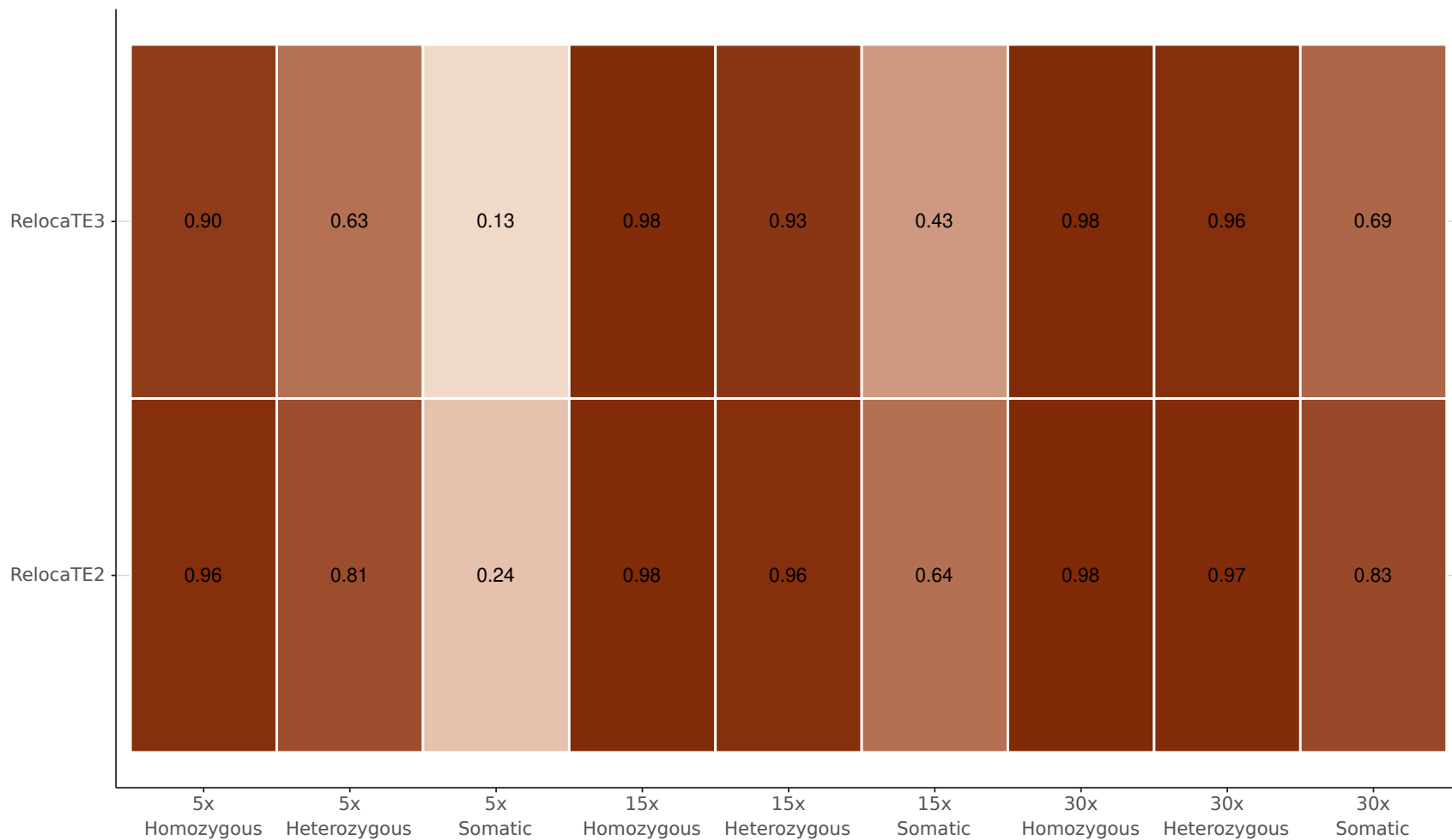
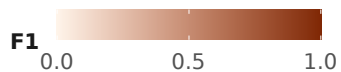
Each row = coverage x class; gap = head-to-head difference

● RelocaTE2 ● RelocaTE3



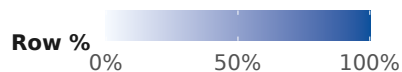
F1 (detection x precision) by condition

Harmonic mean of recall and precision (precision = global-denominator, per sample)



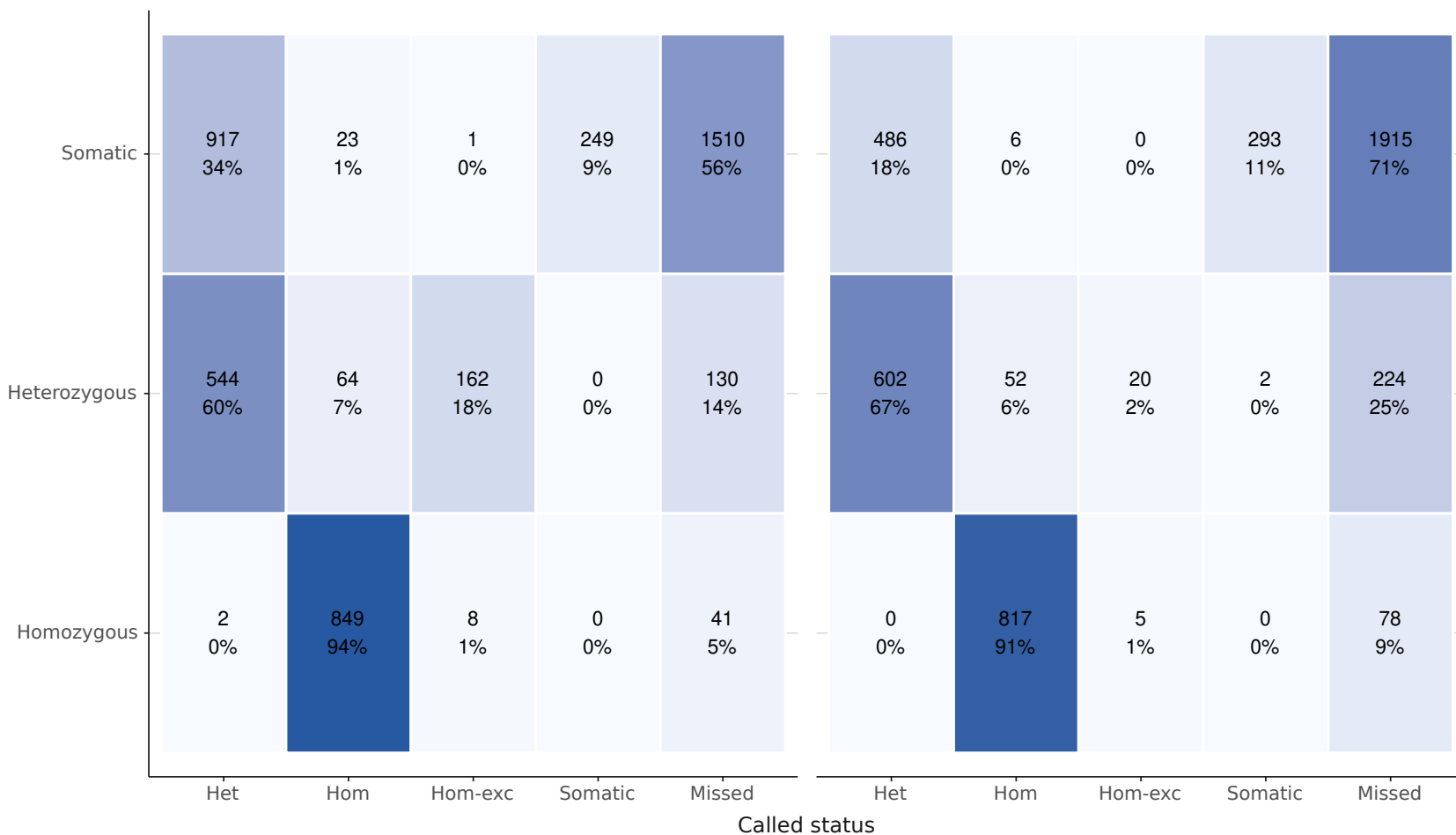
Genotype / detection confusion

Row-normalised: where each truth class's events land (pooled over coverage)



RelocaTE2

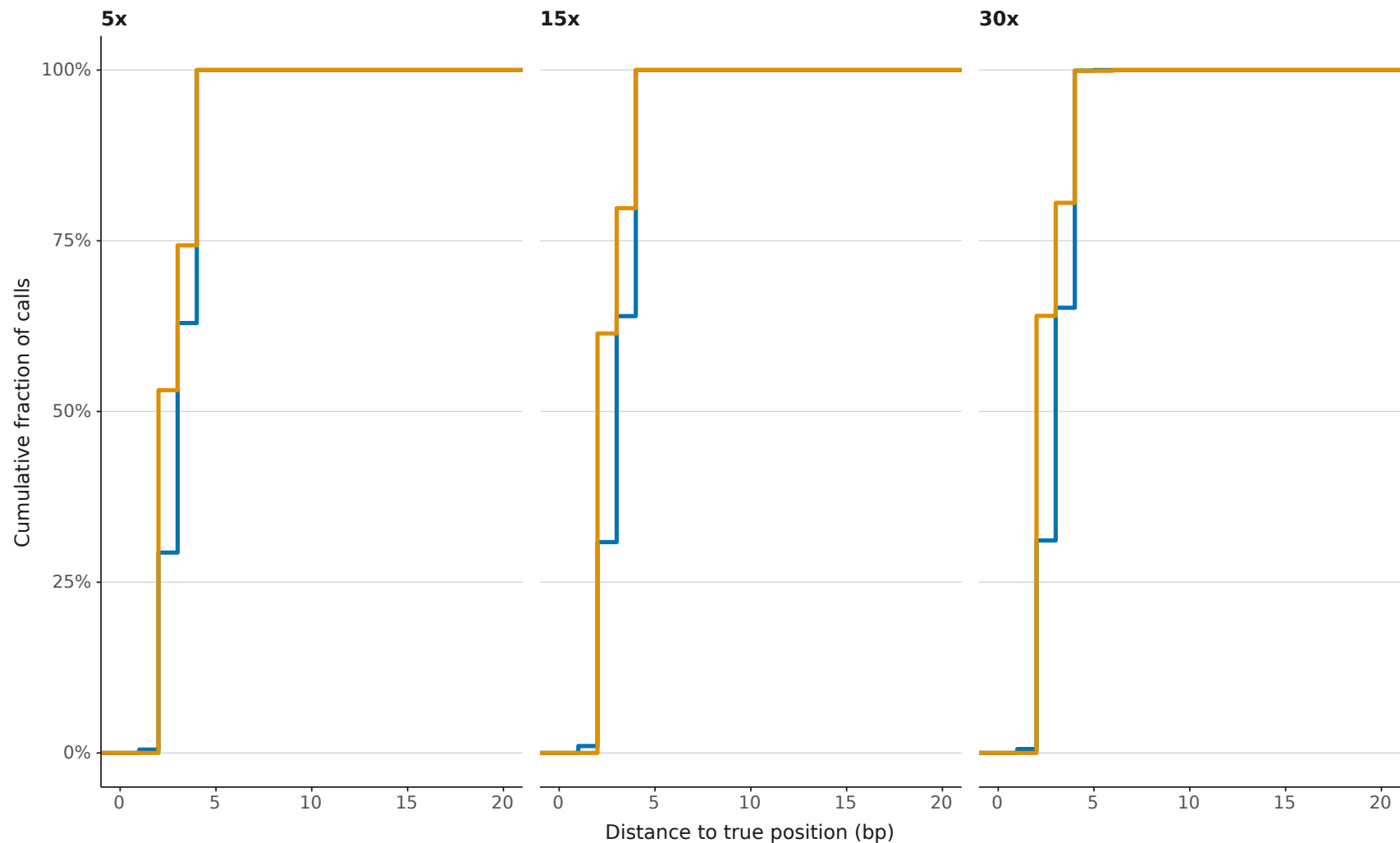
RelocaTE3



Breakpoint positional accuracy

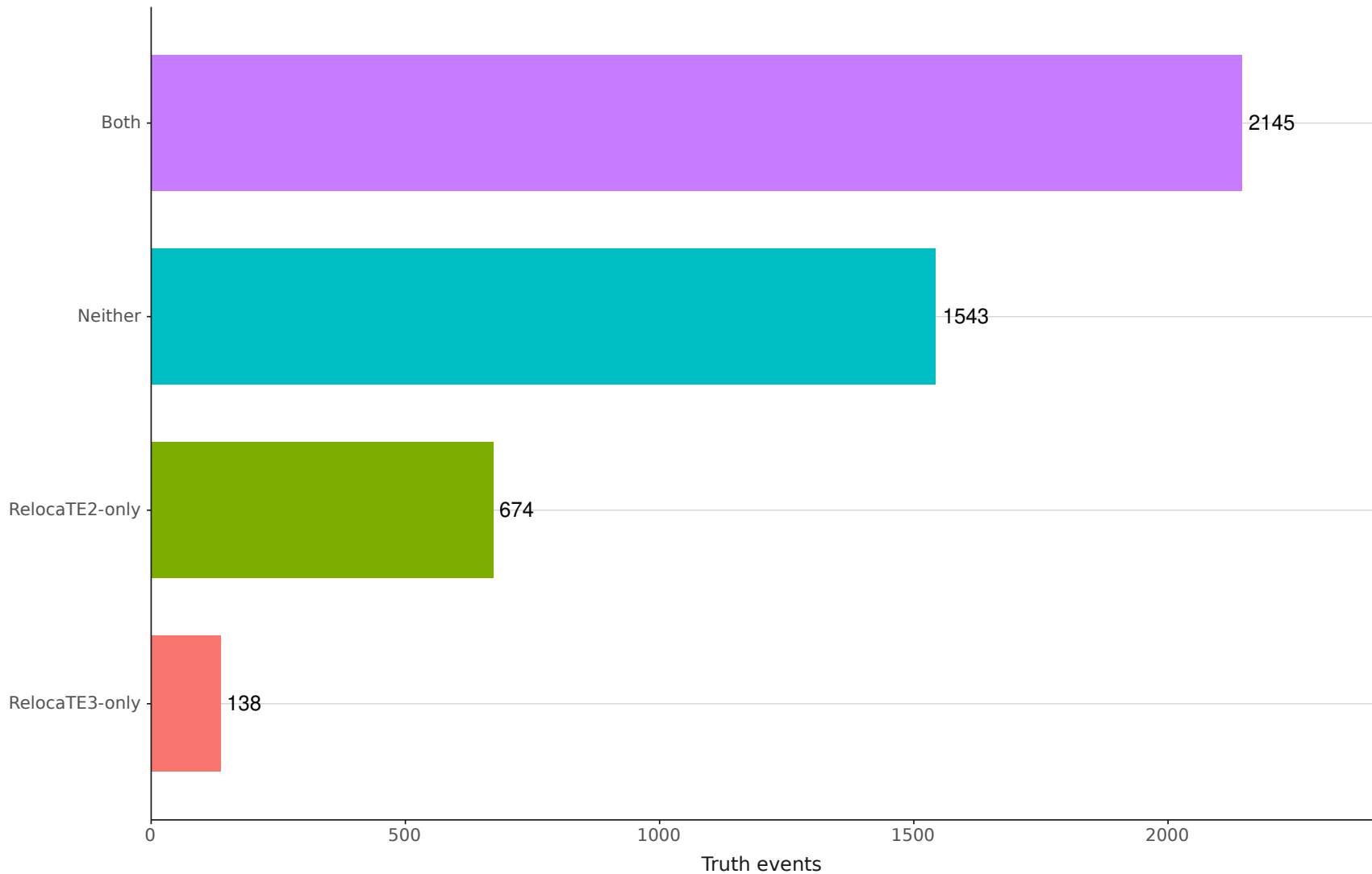
ECDF of $|\text{called} - \text{true}|$ position for matched events; steeper/left = better

RelocaTE2 RelocaTE3



Per-event caller agreement

Truth events by which caller(s) detected them (pooled over all samples)



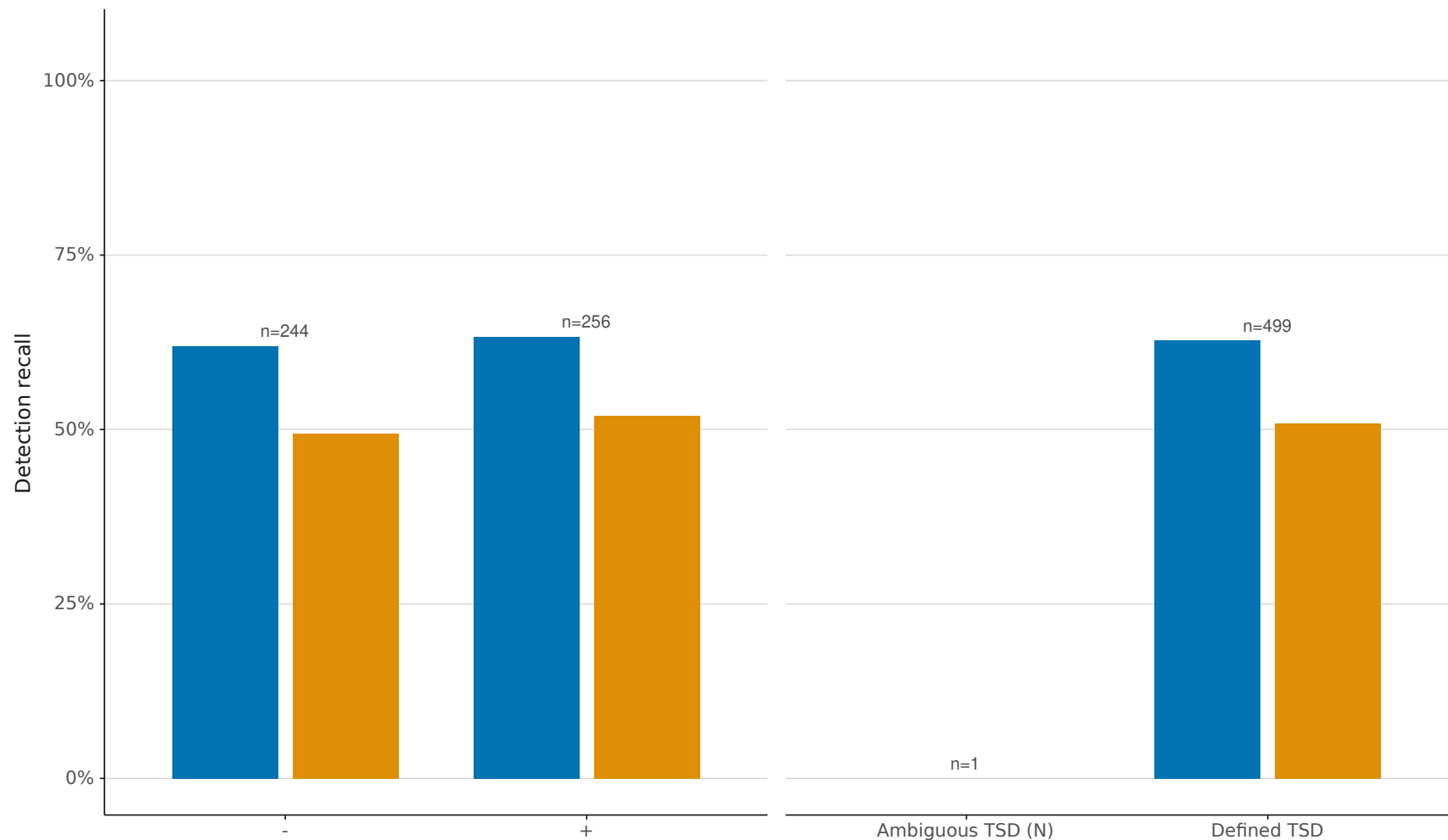
What do missed events have in common?

Detection recall stratified by strand and TSD ambiguity (pooled); n = distinct truth events

RelocaTE2 RelocaTE3

Strand

TSD definition



Compute resources vs. coverage

Per-sample wall time and peak RSS (mean $\pm$ SEM)

